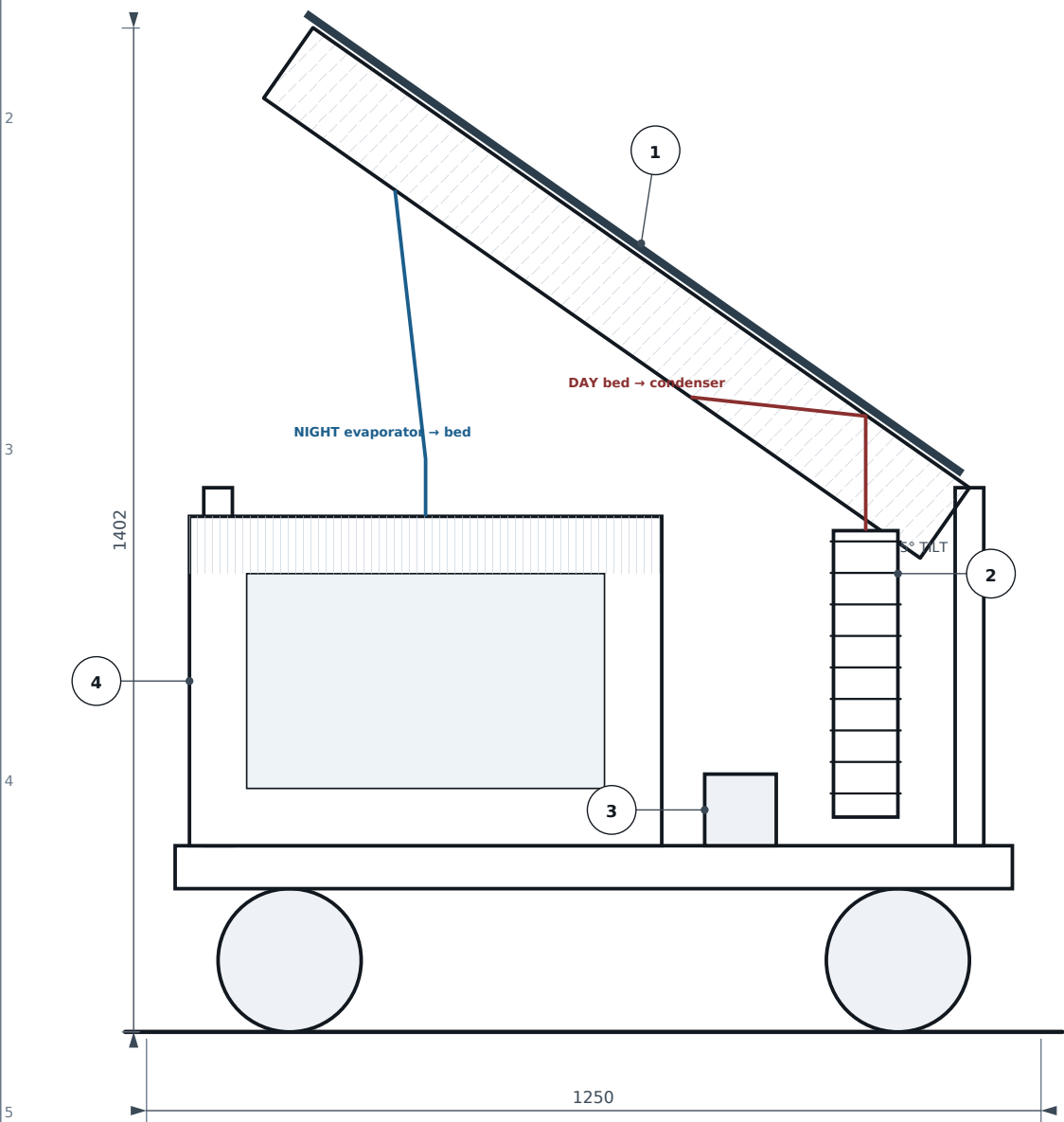


CONTAINS 4.9 kg OF METHANOL — TOXIC AND FLAMMABLE — OUTDOOR INSTALLATION ONLY, SEE SI-300

SIDE ELEVATION

SCALE 1:10

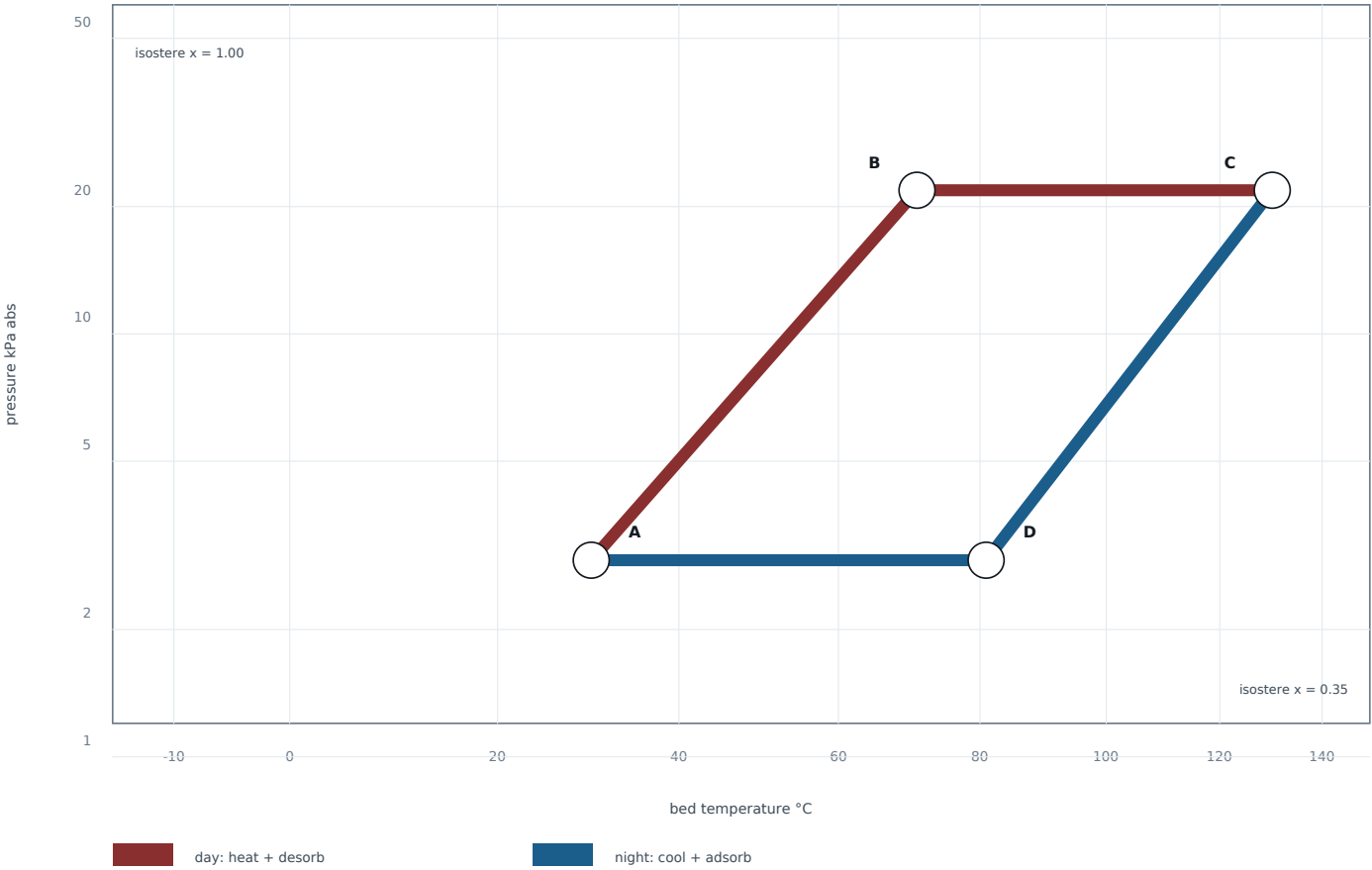


TOTAL MASS 91 kg CHARGED — a wheeled trolley, not a carried unit

MASS		
adsorber tubes, SS 316L		17.5 kg
absorber plate, Cu 0.5		4.5 kg
glazing, low-iron 3.2		8.0 kg
collector box + insulation		12.0 kg
activated carbon		4.9 kg
METHANOL CHARGE		4.9 kg
condenser + receiver		9.0 kg
evaporator + ice box		14.0 kg
frame, wheels, handle		16.0 kg
TOTAL, CHARGED		90.8 kg

CLAPEYRON CYCLE

ln P vs -1/T



BILL OF MATERIALS

ITEM	PART	DESCRIPTION	QTY
1	SI-101	ADSORBER-COLLECTOR, 8 TUBE	1
2	SI-102	CONDENSER, FINNED, NATURAL CONVECTION	1
3	SI-103	RECEIVER, 3 L	1
4	SI-104	EVAPORATOR + ICE BOX	1
5	SI-105	CHECK VALVE, ALL-METAL	2
6	—	ACTIVATED CARBON, MAXSORB III	4.9 kg
7	—	METHANOL, ANHYDROUS >99.8 %	4.9 kg

INDIA-VC THERMAL HARDWARE

ISO 128 / ISO 2768-mK

TITLE

SOLAR ICE MAKER — ARRANGEMENT

ACTIVATED CARBON + METHANOL / 1.0 m2 / 5 kg ICE PER DAY / OUTDOOR ONLY

SCALE

1:10

UNITS

mm

THIRD ANGLE PROJECTION

DRAWING No.

SI-100

REV

A

SHEET

1 / 6

DRAWN

AI-ASSISTED DRAFT

CHECKED

— PENDING —

MATERIAL

SEE BOM

FINISH

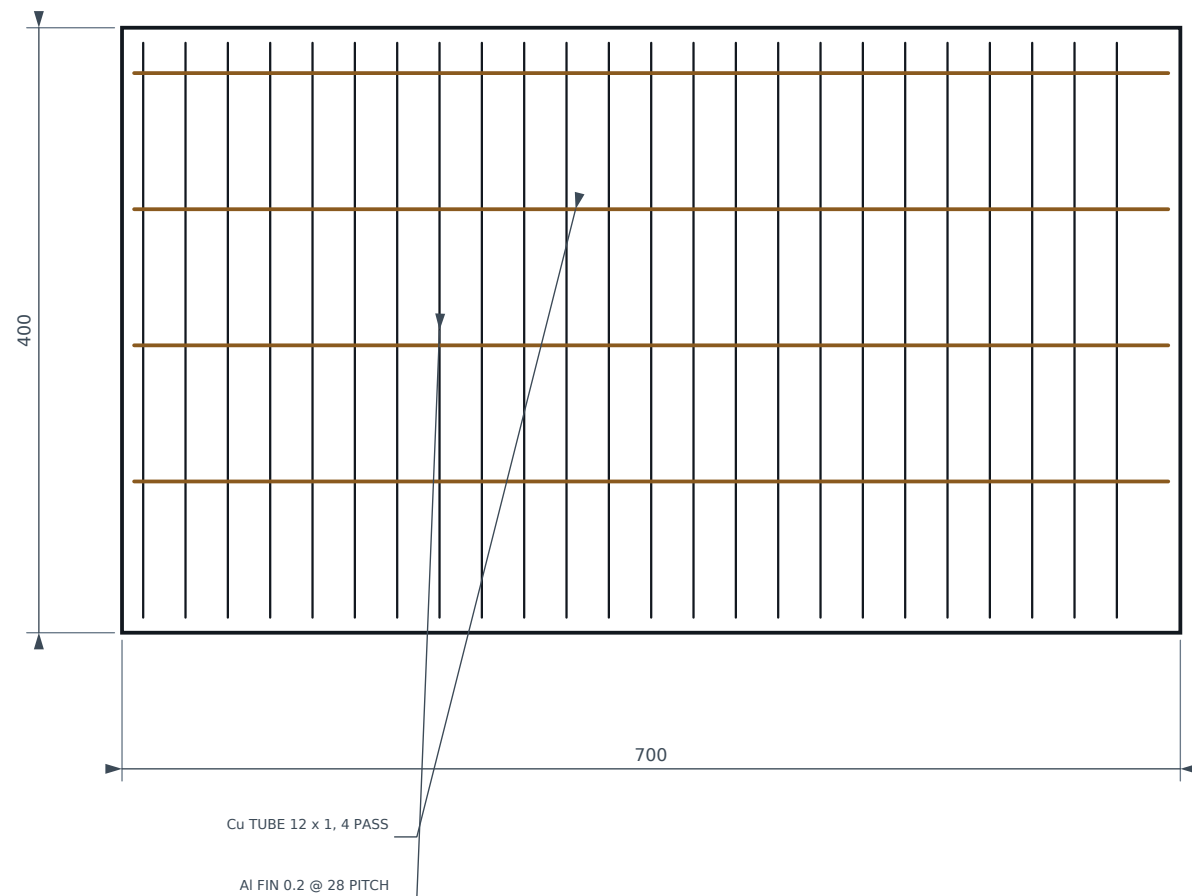
SEE NOTES

DATE

2026-09-15

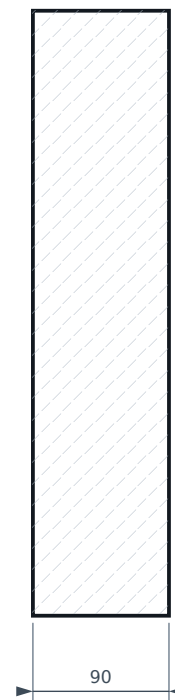
CONDENSER – FRONT

SCALE 1:5



SIDE

1:5

**RECEIVED SI-103**

Ø100

CONDENSER AND RECEIVER DATA			
condensing temperature			30 °C
condensing pressure			21.9 kPa abs
duty per day			3.76 MJ
mean power over 6 h			174 W
air-side dT			15 K
h, convection + radiation			10 W/m²K
surface required			1.16 m²
methanol condensed per day			3.19 kg
receiver working volume	0		3.0 L

NOTES

1. No fan. The condenser works on buoyancy and radiation, so it must be shaded from the sun and free to draw air from below.
2. Mount on the shaded side of the collector with 150 mm clear beneath and above.
3. Receiver sits below the condenser and above the evaporator: the loop is gravity-driven, there is no pump.
4. Cu tube to SS header joints brazed with Ag, then the whole assembly TIG welded into the envelope.
5. Fin pitch 28 is coarse on purpose. Natural convection chokes below about 6 mm free spacing.
6. Receiver has no sight glass and no drain. The charge is not serviceable in the field.

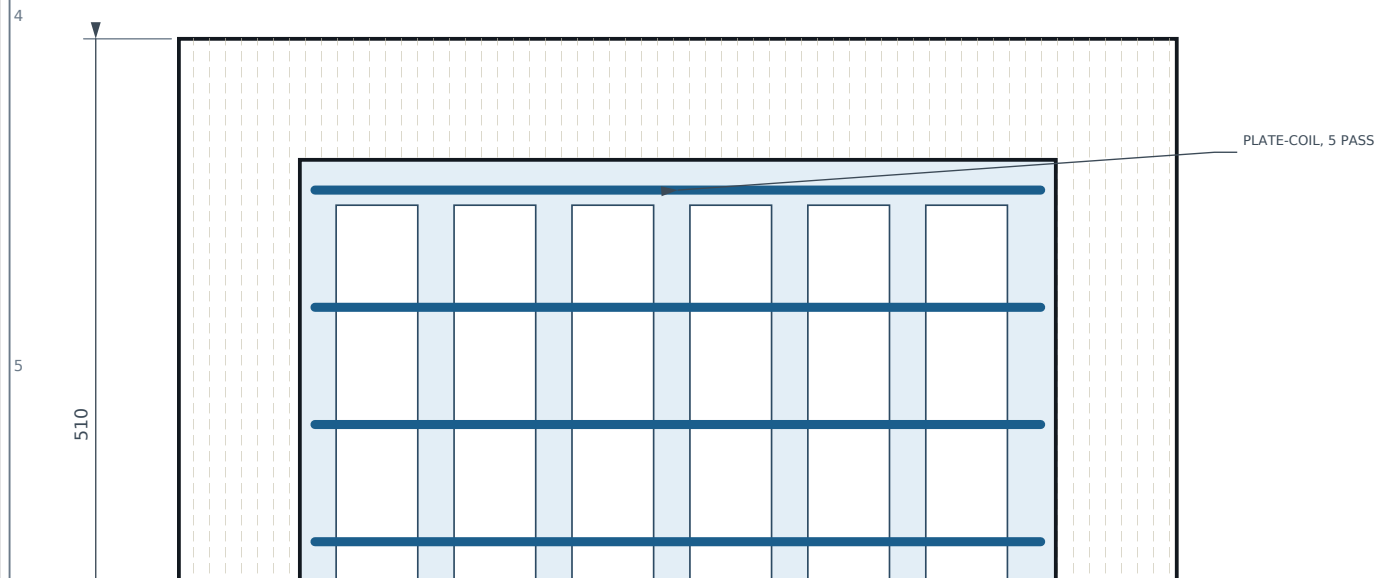
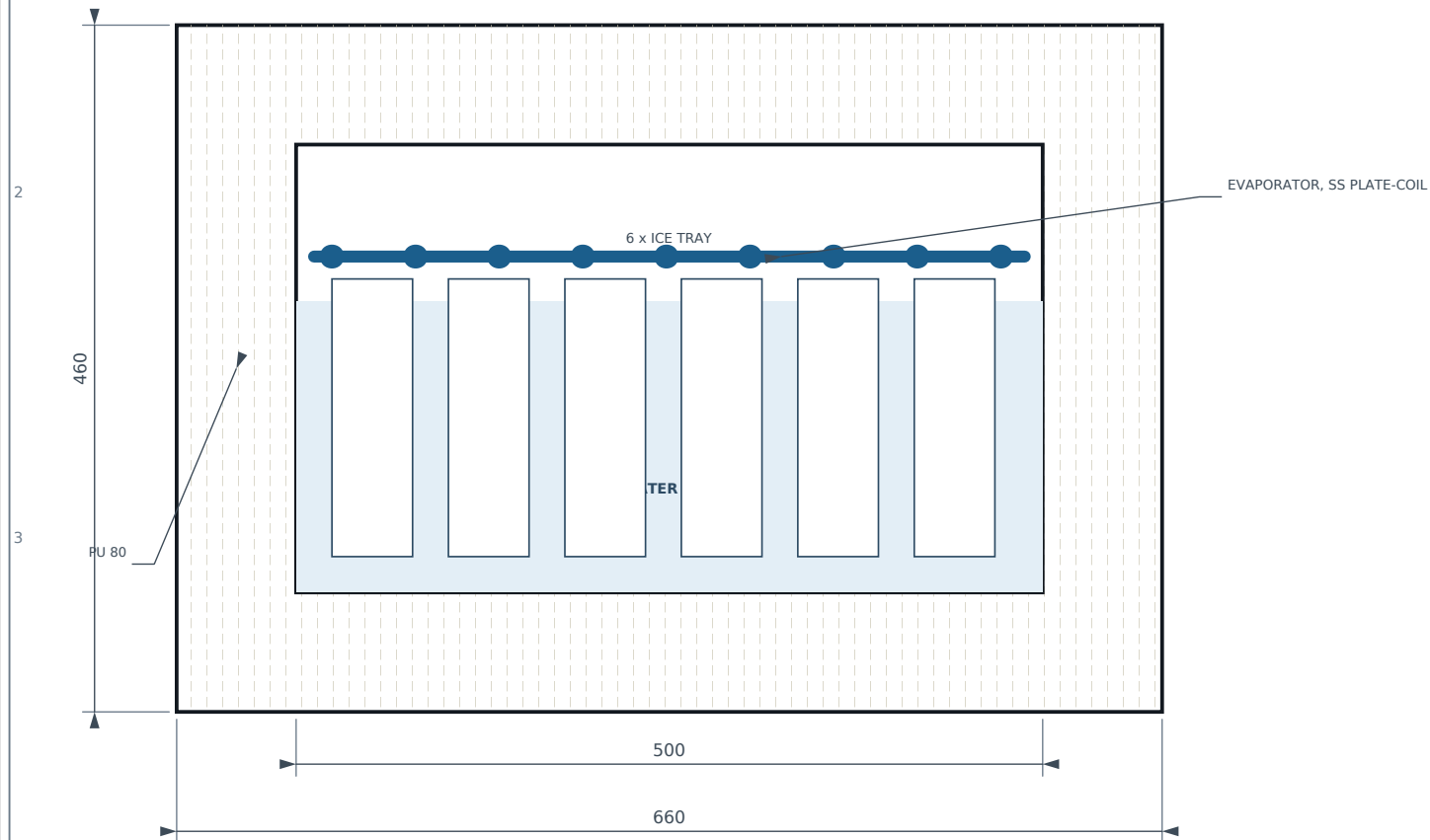
WORKING LEVEL 3.2 kg

INSPECTION		
CHECK	REQUIREMENT	METHOD
Helium leak, whole assembly	< 1e-8 Pa.m3/s	mass spectrometer
Proof pressure	21 bar, 5 min	hydrostatic, then dry
Condenser orientation	tubes horizontal, drain to receiver	level check
Shading	no direct sun on the fins	site survey

INDIA-VC THERMAL HARDWARE					ISO 128 / ISO 2768-mK	
TITLE CONDENSER AND RECEIVER AIR-COOLED, NATURAL CONVECTION / 174 W MEAN / 1.16 m2			SCALE 1:5 THIRD ANGLE PROJECTION		UNITS mm	
DRAWING No. SI-102 / SI-103			REV A		SHEET 3 / 6	
DRAWN AI-ASSISTED DRAFT	CHECKED — PENDING —	MATERIAL SEE BOM	FINISH SEE NOTES		DATE 2026-09-15	

ICE BOX – SECTION

SCALE 1:5



INSPECTION		
CHECK	REQUIREMENT	METHOD
Evaporator external collapse	full vacuum, no set	1 bar external, 1 h
Helium leak	< 1e-8 Pa.m3/s	mass spectrometer
Containment tray	welded, drains to bund	water fill test
Box heat leak	<= 10 W at 30 C	calibrated pull-down

EVAPORATOR AND BOX DATA	
evaporating temperature	-5 C
evaporating pressure	2.92 kPa abs
cooling delivered per night	2.19 MJ
mean power over 12 h	51 W
surface required	508 cm ²
box internal	500 x 350 x 300
insulation U	0.30 W/m ² K
heat leak at 30 C ambient	8 W
leak as share of the night	15 %

NOTES

1. The evaporator runs at 2.9 kPa absolute. It sees full external atmospheric pressure whenever the machine is cold - design it as a vacuum vessel, not a pressure vessel.
2. Plate-coil in SS 316L, all welded, no fittings inside the box.
3. Evaporator sits ABOVE the trays so liquid methanol drains back to the receiver by gravity when the machine warms.
4. Secondary containment: the box is lined with a welded SS tray connected to the bund on SI-300. A leak must not reach the ice.
5. The ice is NOT potable. Methanol is on the other side of one wall. Use the ice for cooling, not for drinks.
6. 80 mm PU gives a 15 % overnight leak, already inside the 5 kg figure.

INDIA-VC THERMAL HARDWARE					ISO 128 / ISO 2768-mK	
TITLE			SCALE		UNITS	
EVAPORATOR AND ICE BOX			1:5		mm	
5 kg ICE PER NIGHT / 51 W MEAN / 80 mm PU			THIRD ANGLE PROJECTION			
DRAWING No.			REV		SHEET	
SI-104			A		4 / 6	
DRAWN	CHECKED	MATERIAL	FINISH		DATE	
AI-ASSISTED DRAFT	— PENDING —	SEE BOM	SEE NOTES		2026-09-15	

1

CHARGING IS THE MOST DANGEROUS OPERATION IN THE LIFE OF THIS MACHINE — OUTDOORS, TRAINED OPERATOR, NO IGNITION SOURCE

CHARGING SEQUENCE

1

WELD OUT

Complete every wetted joint. No threaded or elastomer joint remains.

2

PROOF PRESSURE

Hydrostatic 21 bar, 5 min. Drain, dry with warm dry nitrogen.

1.5 x stagnation

3

LEAK TEST

Helium mass spectrometer under vacuum on the whole envelope.

< 1e-8 Pa.m3/s

4

BAKE-OUT

200 C / 8 h at < 1 Pa through the charge tube, pumping throughout.

carbon outgasses

5

COOL UNDER VACUUM

Let the bed reach ambient still under the pump. Do not vent.

no air, ever

6

CHARGE

Admit 4.9 kg anhydrous methanol by WEIGHT from a sealed burette.

by mass, not volume

7

DEGAS

Freeze-pump-thaw x3 on the receiver. Final pump to < 50 Pa at 0 C.

removes dissolved air

8

PINCH-OFF

Cold-weld the charge tube, cut, then TIG cap the stub.

permanent

9

LEAK TEST 2

Helium bomb 3 bar / 4 h, then mass spectrometer.

< 1e-8 Pa.m3/s

10

FIRST CYCLE

One full solar cycle under observation. Record bed and receiver temperatures.

commissioning

CHARGING RIG

Schematic

METHANOL BURETTE

COLD TRAP LN2 / dry ice

VACUUM PUMP 2-stage rotary

SI-100 MACHINE

The cold trap protects the pump and, more importantly, stops methanol vapour reaching the exhaust. Vent the pump outdoors, downwind.

CHARGING PARAMETERS		
PARAMETER	VALUE	LIMIT
Methanol charge	4.91 kg	by weight, ±1 %
Purity	anhydrous > 99.8 %	water poisons the carbon
Bake-out	200 C / 8 h	< 1 Pa
Final vacuum before seal	< 50 Pa	at 0 C
Proof pressure	21 bar	1.5 x stagnation
Leak rate	< 1e-8 Pa.m3/s	helium, twice
Seal	cold-weld + TIG cap	no valve, no port

PROHIBITED

Venting the envelope to air at any point after bake-out. Air in the bed is non-condensable gas and it never comes out again.

Charging by volume. Methanol expands 1.2 % per 10 K; the charge is a mass specification.

Any valve, gauge port or sight glass left on the wetted envelope. Every one is a leak path for the life of the machine.

Charging indoors, or anywhere without free ventilation and no ignition source within 5 m.

INDIA-VC THERMAL HARDWARE

ISO 128 / ISO 2768-mK

TITLE

VACUUM AND CHARGING

4.9 kg METHANOL / ONE-TIME PERMANENT CHARGE / NO FIELD SERVICE

SCALE

NONE

THIRD ANGLE PROJECTION

UNITS

mm

DRAWING No.

SI-200

REV

A

SHEET

5 / 6

DRAWN

AI-ASSISTED DRAFT

CHECKED

— PENDING —

MATERIAL

SEE BOM

FINISH

SEE NOTES

DATE

2026-09-15

1

THIS IS NOT A CERTIFIABLE DOMESTIC APPLIANCE

Methanol has no ASHRAE refrigerant designation for this service and grades as both toxic and highly flammable. There is no path to IEC 60335-2-40 approval for indoor use.

HAZARD DATA		
PROPERTY	VALUE	CONSEQUENCE
Charge	4.9 kg methanol	outdoor only
Flash point	11 C closed cup	below ambient
Flammable range	6 – 36 % in air	unusually wide
Flame	nearly invisible in daylight	cannot be seen
Acute toxicity	10 mL blinds, 30 mL can kill	skin absorbs it
Occupational limit	200 ppm TWA	low
Operating pressure	8.4 bar at 130 C	vessel, not plumbing
Stagnation pressure	14.0 bar at 150 C	the design case

DESIGN CONSEQUENCES ALREADY BUILT IN

- All-welded SS 316L and copper envelope. No elastomer, no thread, no service port anywhere wetted.

- One-time permanent charge, sealed by cold-weld pinch-off and a welded cap. The machine is replaced, never opened.

- No pressure relief to atmosphere. The tubular adsorber contains stagnation at 4.9 x margin instead of venting methanol.

- Welded stainless containment tray under the evaporator, receiver and condenser, drained to a bund sized for the whole charge.

- Ice box lined and sealed: a leak cannot reach the ice. The ice is not potable in any case.

REQUIRED MARKING

ON THE VESSEL AND THE FRAME

!

METHANOL CH3OH

4.9 kg — TOXIC, FLAMMABLE

INVISIBLE FLAME

OUTDOOR USE ONLY · DO NOT OPEN · DO NOT VENT

ICE NOT POTABLE · NO IGNITION SOURCE WITHIN 5 m

COMMISSIONING		
STEP	COMMISSIONING	RECORD
1	Site: outdoors, level, free air, 5 m from ignition	site sheet
2	Confirm bund and tray drain clear	visual
3	Leak test 2 witnessed before first sun	He, < 1e-8
4	First solar cycle observed end to end	T bed, T receiver
5	Confirm condenser drains fully to the receiver	no hold-up
6	Measure ice after night one	kg, vs 5 kg target
7	Record stagnation temperature on a no-load day	T max

IF THE MACHINE IS DAMAGED

- Do not approach a suspected leak with any ignition source. The flame is invisible in daylight; use a thermal camera or a broom handle.

- Methanol vapour is heavier than air near the source and pools in the bund. Ventilate from upwind.

- Do not attempt to recover the charge. Isolate the machine, let it weather in open air, and dispose of the carbon and vessel through a chemical waste route.

- A machine that has been vented to atmosphere is finished as a machine: air in the bed cannot be pumped out in the field.

MACHINE SUMMARY	
Designed output	5 kg ice / day
Solar COP	0.110
Collector	1.0 m2
Total mass, charged	91 kg
Certification status	NONE — demonstrator

INDIA-VC THERMAL HARDWARE

ISO 128 / ISO 2768-mK

TITLE		SCALE	UNITS	
SAFETY, MARKING AND COMMISSIONING		NONE	mm	
OUTDOOR OFF-GRID EQUIPMENT / 4.9 kg METHANOL / NOT A DOMESTIC APPLIANCE		THIRD ANGLE PROJECTION		
DRAWING No. SI-300		REV A	SHEET 6 / 6	
DRAWN AI-ASSISTED DRAFT	CHECKED — PENDING —	MATERIAL SEE BOM	FINISH SEE NOTES	DATE 2026-09-15